Project 1: Social Media Engagement & Sentiment Analysis

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Key results

1. TikTok and Instagram posts get more likes than LinkedIn and Facebook posts: median likes TikTok 44, Instagram 37, Twitter 32, Facebook 26, LinkedIn 20.
2. The ranking holds after adjusting for audience size: median engagement per follower is TikTok 0.0346 vs LinkedIn 0.0124.

1. Dataset, source and provenance

Source: GitHub repository mrudukp/social-media-sentiment-analysis (https://github.com/mrudukp/social-media-sentiment-analysis), file social_media_sentiment_raw.csv. License: not stated. Collection method: not stated; possibly synthetic (the text column has only 150 unique values across 2,200 posts). Findings describe this dataset only, not any real brand or country. One record = one social media post by one user on one platform at one timestamp.

2. Data checks

Raw file: 2,260 rows and 13 columns. After cleaning: 2,200 rows.

The cleaning log lists 11 steps: converting followers_count, likes, retweets_shares and comments to numbers; setting negative counts to missing; parsing timestamps; dropping exact duplicate rows; and dropping repeated post_id (keeping the first). The log records 0 rows or cells affected at every step, so it does not explain the 60 rows removed between the raw file (2,260) and the cleaned file (2,200).

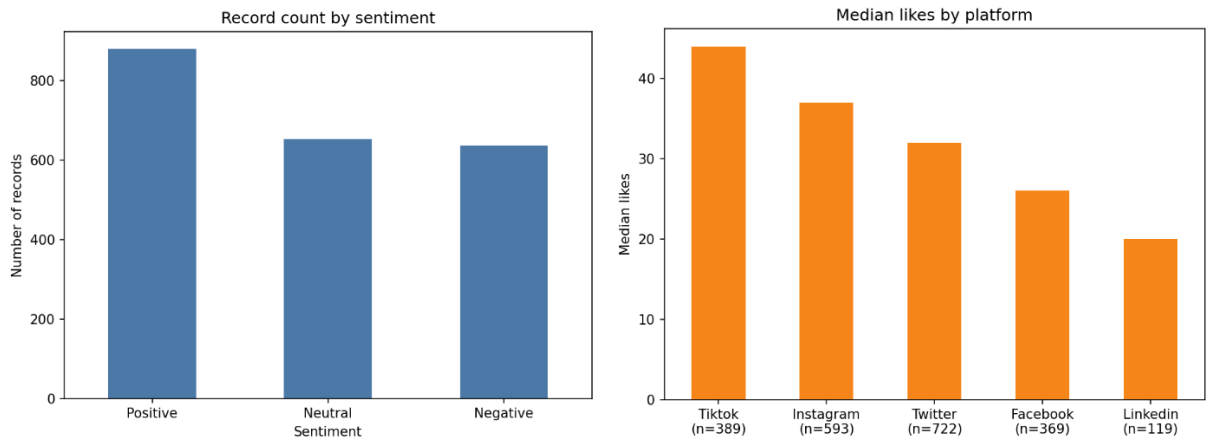
Missing values after cleaning: hashtags 762 (probably posts with no hashtags), followers_count 88, country 66, comments 44, verified 44, sentiment 33, likes 8. Missing values were reported, not filled or dropped.

3. Summary statistics

Overall (n = 2,200): median likes 33 (mean 36.1, 25th to 75th percentile 25 to 44, max 126); median shares 6; median comments 3; median followers 1,800 (max 234,473, very skewed). The top 10% of posts account for 20% of all likes, so a few posts do not dominate the average. Likes and comments have a moderate Spearman correlation of 0.59.

Sentiment counts: Positive 879, Neutral 652, Negative 636 (33 missing).

Platform	Posts	Median likes
TikTok	389	44
Instagram	593	37
Twitter	722	32
Facebook	369	26
LinkedIn	119	20



4. Findings

Finding 1: TikTok and Instagram posts get more likes than LinkedIn and Facebook posts. Median likes: TikTok 44 (n = 389), Instagram 37 (n = 593), Twitter 32 (n = 722), Facebook 26 (n = 369), LinkedIn 20 (n = 119). Median comments follow the same order. The TikTok to LinkedIn gap is more than double.

Finding 2: The ranking holds after adjusting for audience size. Median engagement per follower, (likes + shares + comments) / followers: TikTok 0.0346 (n = 379), Instagram 0.0287 (n = 567), Twitter 0.0210 (n = 692), Facebook 0.0159 (n = 362), LinkedIn 0.0124 (n = 112). The Finding 1 gap is not explained by audience size alone.

5. Marketing recommendation

In this dataset, TikTok and Instagram posts have higher median likes and higher engagement per follower than LinkedIn and Facebook posts. The brand could run a small controlled test, posting comparable content on TikTok, Instagram and LinkedIn for the same period, and compare results using one consistent metric such as median engagement per follower.